Universal Radiative Corrections for Precision Determinations of V_{ud}

BEACH conference

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1. Semi-leptonic kaon decays
2. Ratio of V_{us}/V_{ud}
3. Radiative corrections and EFT formalism
4. NLO EW corrections
5. NNLO EW corrections

- SM motivation - split theory and exp
- define acronims
- decays Kl3 - feyn diags as ex
- more info in plots context
- smoth doen transition to my computations

- numbers off eq
- move definition of EFT before pion and neutron
- explain the calculations that have been done by martin and francesno at NNLOQCD/EW
- talk about precision of Vud and is very precise with these new calculation but EW corrections can improve uncertainty which we are doing and use this to motivate our work
- take off 1 loop diagrams
- add tadpole box on 2 loop
- talk about tadpoles and how they contribute
- present C in fixed scheme

- CKM (mixing of quark flavour eigenstates)
- Cabbibo angle anomaly;
- Measure decay rates: K, $0^+ \longrightarrow 0^+$, Π, neutron decay;
- K_{l3}, $0^+ \longleftarrow 0^+$, neutron decay (SL) and K_{l2} (L);
- Status of V_{us} : Kaon decays and status of V_{ud} : superallowed β decays, neutron decay, pion decays;

Lattice constrains

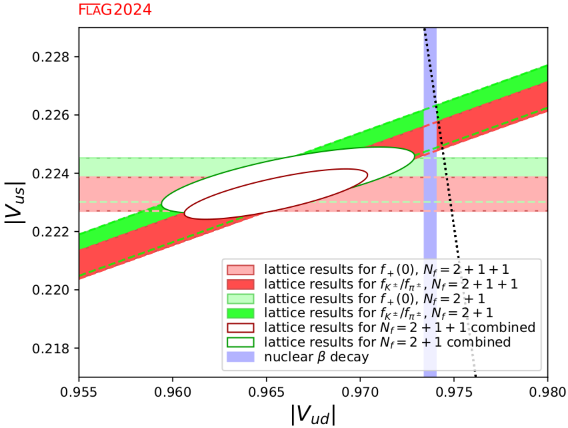


Figure 1: V_{ud} and V_{ud} using lattice QCD input and V_{ud} extracted from nuclear β decays.

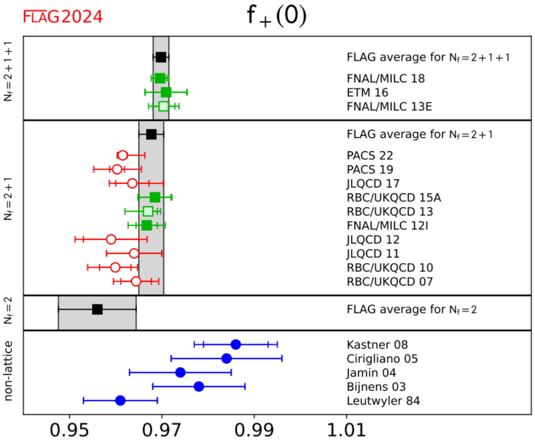
$$\Gamma(K_{\ell 3}) = \frac{G_F^2 M_K^5}{192\pi^3} C_K^2 \, S_{\mathrm{EW}} \, |V_{us}|^2 \left| f_+^{K^0\pi^-}(0) \right|^2 I_{K\ell}^{(0)} \left(1 + \delta_{SU(2)}^{K\pi} + \delta_{\mathrm{EM}}^{K\ell}\right)^2$$

- $S_{EW} = 1.0232(3) \longrightarrow$ process indepented EW corrections;
- $I_{K\ell}^{(0)}$ phase space in isospin limit;
- $\delta_{SU(2)}^{K\pi}$ strong isospin breakig correction
- $\delta_{\mathrm{EM}}^{K\ell}$ long distance EW corrections

$$\langle \pi(p_\pi) | \bar{s}\gamma^\mu u | K(p_K) \rangle = (p_K + p_\pi)^\mu f_+(q^2) + (p_K - p_\pi)^\mu f_-(q^2), \quad (1)$$

Experimental inputs

$$|V_{us}| \, f_+(0) = 0.21654(41)$$



$$f_+(0) = 0.9698(17)$$

$$|V_{us}| = 0.22328(58) \qquad (K_{\ell 3})$$

Ratio of V_{us}/V_{ud}

$$\frac{\Gamma(K^{\pm}_{\mu 2(\gamma)})}{\Gamma(\pi^{\pm}_{\mu 2(\gamma)})} = \left|\frac{V_{us}}{V_{ud}}\right|^2 \left(\frac{f_{K^\pm}}{f_{\pi^\pm}}\right)^2 \frac{M_{K^\pm}\left(1 - m_\mu^2/M_{K^\pm}^2\right)^2}{M_{\pi^\pm}\left(1 - m_\mu^2/M_{\pi^\pm}^2\right)^2} (1 + \delta_{\mathrm{EM}})$$

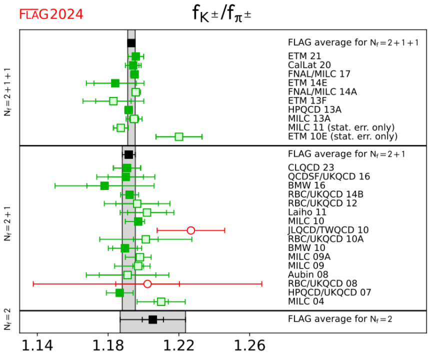
$$\langle 0|\bar{q}_1 \gamma^\mu \gamma_5 q_2|P(p)\rangle = i\, p^\mu f_P, \qquad P = \pi^+,\, K^+. \tag{2}$$

- Average of $f_K/f_P i$ in isosymmetric QCD;

Experimental inputs

$$\left|\frac{V_{us}}{V_{ud}}\right| \frac{f_{K^\pm}}{f_{\pi^\pm}} = 0.27599(41)$$

- Isospin correction from the lattice can be improved by higher statistics in contrast with extractions from ChPT;
- Isopsin corrections are becoming relevant for CKM elements extraction at the current level of precision of lattice inputs;



$$|V_{ud}| = 0.97373(31) \qquad \text{(superallowed } \beta \text{ decays)}$$

$$\frac{f_{K^\pm}}{f_{\pi^\pm}} = 1.1934(19)$$

$$\left|\frac{V_{us}}{V_{ud}}\right| = 0.23126(50) \qquad (K_{\ell 2}/\pi_{\ell 2})$$

V_{ud} from pion decay

$$\Gamma_{\pi\beta} = \frac{G_F^2 |V_{ud}|^2 M_\pi^5}{64\pi^3} \left|g_V^\pi\right|^2 \left|f_+^\pi(0)\right|^2 I_{\pi\ell}\left(1 + \Delta I_{\pi\ell}\right) \tag{3}$$

- $f_+^\pi(0)$: pion vector form factor.
- $I_{\pi\ell}$: phase-space integral.
- $\Delta_{\mathrm{RC}}^{\pi\ell}$: radiative corrections.
- g_V^π: effective vector coupling containing the short-distance electroweak corrections.

$$\Delta_{\mathrm{RC}}^{\pi\ell} = 0.03403(11)$$

$$V_{ud}^\pi = 0.97346(283)$$

- The Behrends–Sirlin–Ademollo–Gatto theorem implies

$$f_+^\pi(0) \simeq 1 - 7 \times 10^{-6},$$

making isospin-breaking corrections negligible.

$$\frac{1}{\tau_n} = \frac{G_F^2 |V_{ud}|^2}{2\pi^3}\, m_e^5\, f\, (1 + 3g_A^2)\, (1 + \Delta_R), \tag{4}$$

- f: phase-space factor.
- g_A: axial-vector coupling.
- Δ_R: electroweak radiative corrections.

$$\Delta_R^V = 2.436(16)\%,$$
$$|V_{ud}|_n = 0.97410(41).$$

- First NLL calculation of mixed $O(\alpha\alpha_s^2)$ corrections.
- Reduces theoretical uncertainty in Δ_R^V.
- Improves consistency of first-row CKM unitarity tests.

Operator product expansion

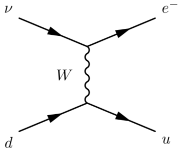


Figure 2: SM tree-level.

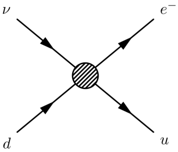


Figure 3: EFT Tree-level.

$$\frac{1}{k^2 - m_W^2} \longrightarrow \frac{-1}{m_W^2}$$

$$H_{eff}(x) = \frac{G_F}{\sqrt{2}} V_{CKM} C(\mu) Q(x)$$

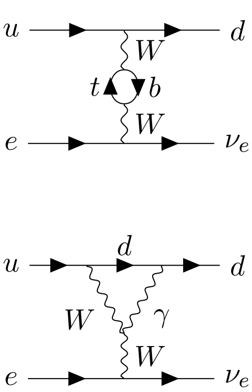


Figure 4: Some 1 loop diagrams in the full theory for the semi-leptonic decay.

Normalization to G_F

$$\mathcal{L}_{G_F} = -\frac{G_F}{\sqrt{2}} Q_\mu - \frac{G_F}{\sqrt{2}} V_{ud} C_q Q_q \tag{5}$$

And the Wilson Coefficients are

$$C_r^{LO} = 1 \tag{6}$$

$$C_r^{NLO} = -5 + \frac{a_\mu^1}{4} + \frac{a_q^1}{12} - 2\log\left(\frac{\mu^2}{M_Z^2}\right) . \tag{7}$$

- Finite, scheme dependent (a_μ^1, a_q^1)
- Only box diagrams contribute

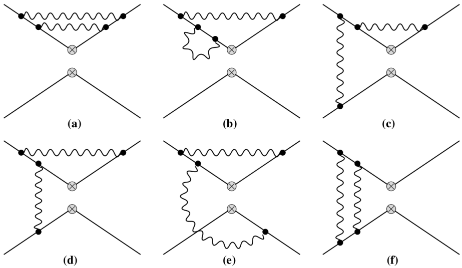


Figure 5: EFT NNLO EW corrections.

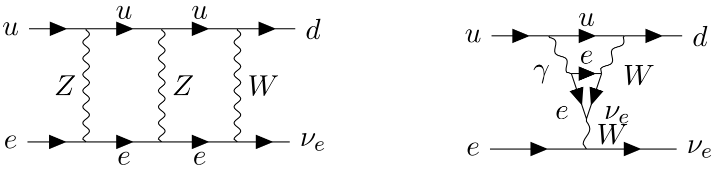


Figure 6: SM NNLO EW corrections.

Conclusions

- Combination of lattice inputs and perturbative calculations;
- Higher order calculation + sub-percent lattice calculations;
- Current ongoing work on NNLO EW corrections for V_{ud} extraction from neutron decay.